Graph-Based Climate Fact Checking

with Neo4j

Ontology-driven claim verification over IPCC documents

Knowledge Graph · Semantic Search · Climate NLP

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1. Motivation

⚠️ Misinformation Crisis

Climate misinformation is spreading rapidly across social media, digital platforms, and political discourse.

📄 Complex Source Material

Scientific climate reports and verifiable evidence are often difficult to navigate and understand, limiting people's ability to critically assess the information they receive.

🔍 Manual Checking Doesn't Scale

Human fact-checkers cannot scale across the volume and complexity of climate claims circulating daily.

🌐 Structured Knowledge Helps

A knowledge graph can represent claims as structured facts, enabling automated, precise comparison and verification.

2. Documents Used

PRIMARY SOURCE

The Intergovernmental Panel on Climate Change (IPCC) publishes comprehensive, peer-reviewed scientific assessment reports every 5-7 years.

IPCC AR6 *Sixth Assessment Report*

- Working Group I — Physical Science Basis
- Working Group II — Impacts & Adaptation
- Working Group III — Mitigation
- Summary for Policymakers (SPM)

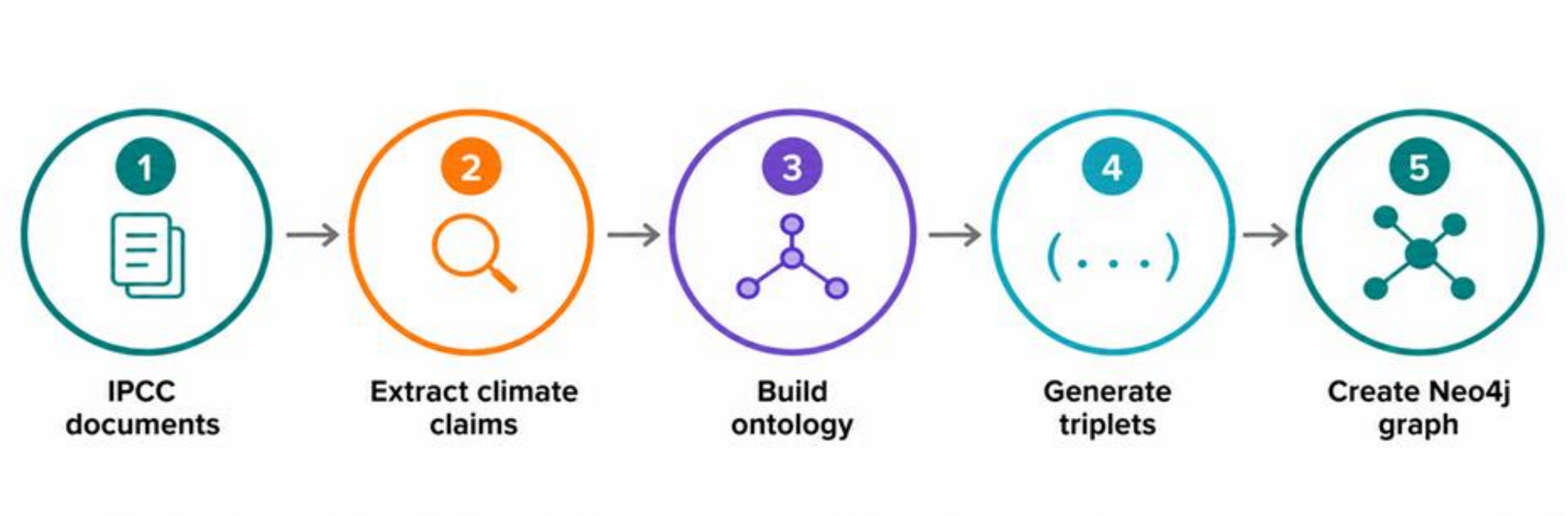
Focus Area

Physical Science Basis - Ocean, Cryosphere and Sea Level Change

Focus Content

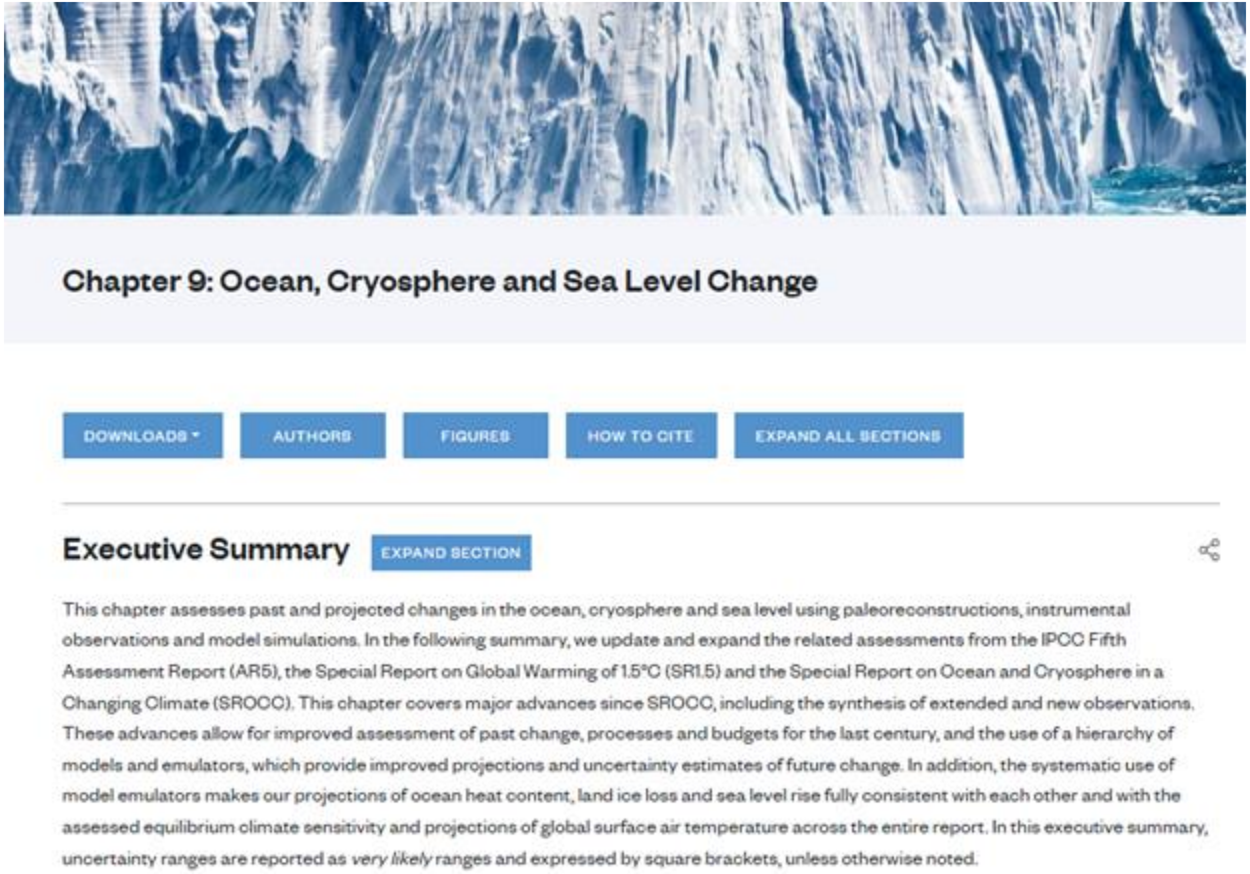
Quantitative climate claims with numerical values

3. Graph Creation — Process



3. Graph Creation — Process

IPCC documents



Chapter 9: Ocean, Cryosphere and Sea Level Change

DOWNLOADS ▾ | AUTHORS | FIGURES | HOW TO CITE | EXPAND ALL SECTIONS

Executive Summary

EXPAND SECTION

This chapter assesses past and projected changes in the ocean, cryosphere and sea level using paleoreconstructions, instrumental observations and model simulations. In the following summary, we update and expand the related assessments from the IPCC Fifth Assessment Report (AR5), the Special Report on Global Warming of 1.5°C (SR1.5) and the Special Report on Ocean and Cryosphere in a Changing Climate (SROCC). This chapter covers major advances since SROCC, including the synthesis of extended and new observations. These advances allow for improved assessment of past change, processes and budgets for the last century, and the use of a hierarchy of models and emulators, which provide improved projections and uncertainty estimates of future change. In addition, the systematic use of model emulators makes our projections of ocean heat content, land ice loss and sea level rise fully consistent with each other and with the assessed equilibrium climate sensitivity and projections of global surface air temperature across the entire report. In this executive summary, uncertainty ranges are reported as *very likely* ranges and expressed by square brackets, unless otherwise noted.

3. Graph Creation — Process

Extract climate claims

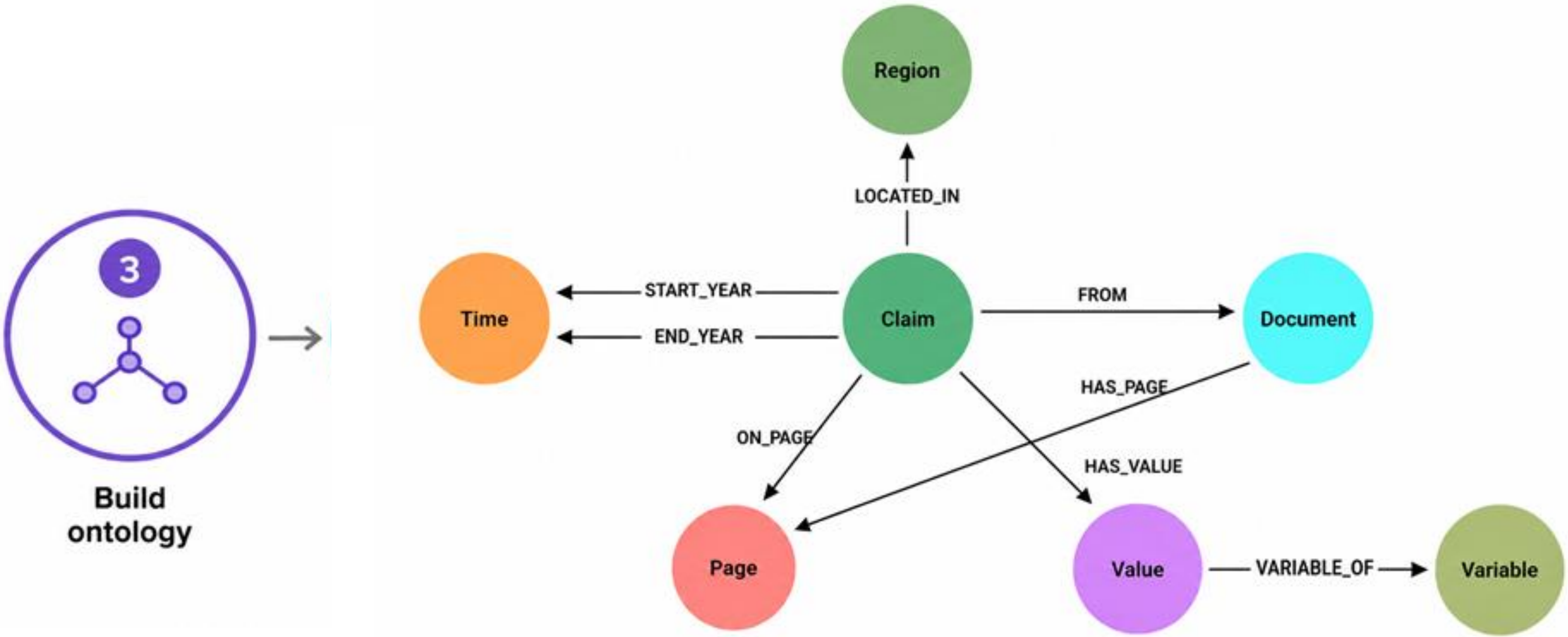
{"claim": **"Increases in greenhouse gas concentrations have led to global mean surface temperature increases of approximately 1 C above pre-industrial levels since the late 19th century."**, "source_pdf": "IPCC_AR6_WGIII_Chapter03.pdf", "source_page": 3, "chunk_id": 15}

{"claim": **"Arctic sea ice extent has decreased by about 13% per decade from 1979 to present."**, "source_pdf": "IPCC_AR6_WGIII_Chapter03.pdf", "source_page": 3, "chunk_id": 15}

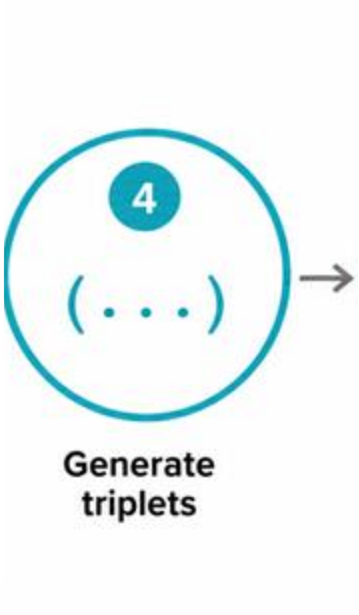
{"claim": **"Global average sea level rise was observed at a rate of around 3.2 mm per year between 2006 and 2015."**, "source_pdf": "IPCC_AR6_WGIII_Chapter03.pdf", "source_page": 3, "chunk_id": 15}

{"claim": **"Global mean surface temperature increased by 1.1C above pre-industrial levels by 2011-2020."**, "source_pdf": "IPCC_AR6_WGIII_Chapter03.pdf", "source_page": 4, "chunk_id": 22}

3. Graph Creation — Process



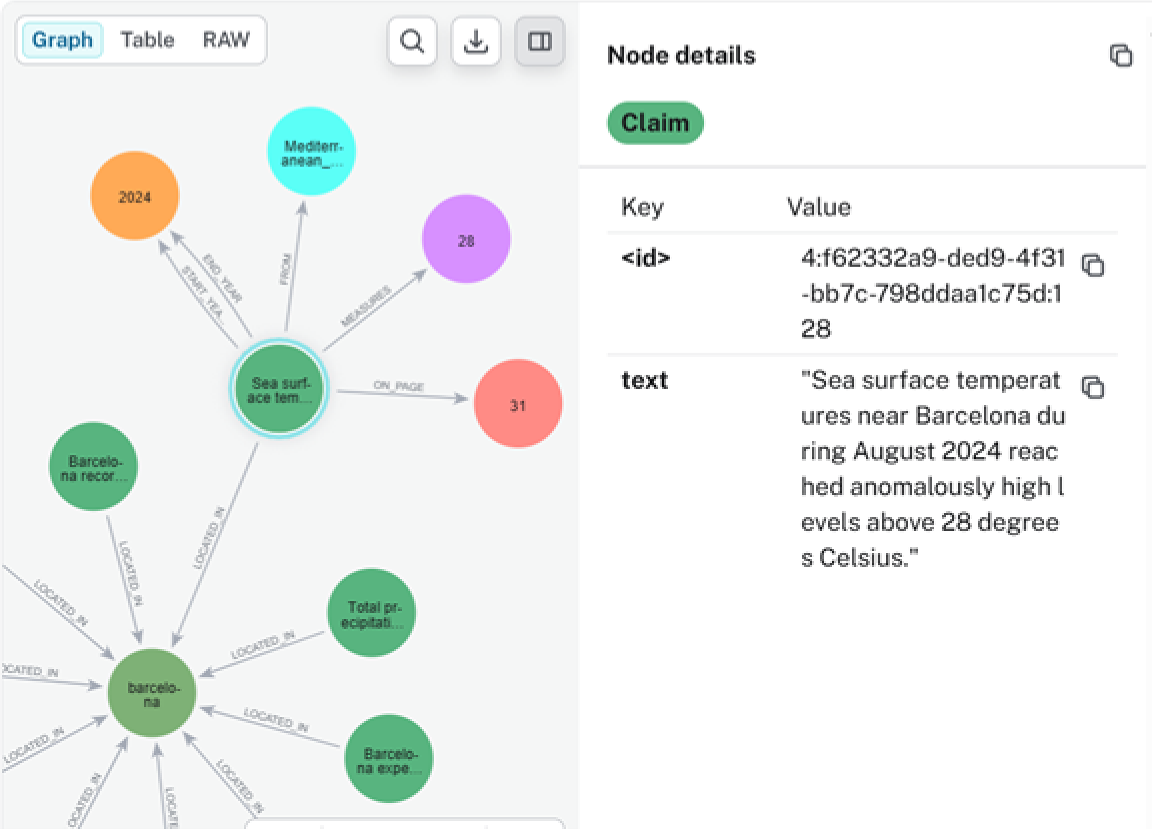
3. Graph Creation — Process



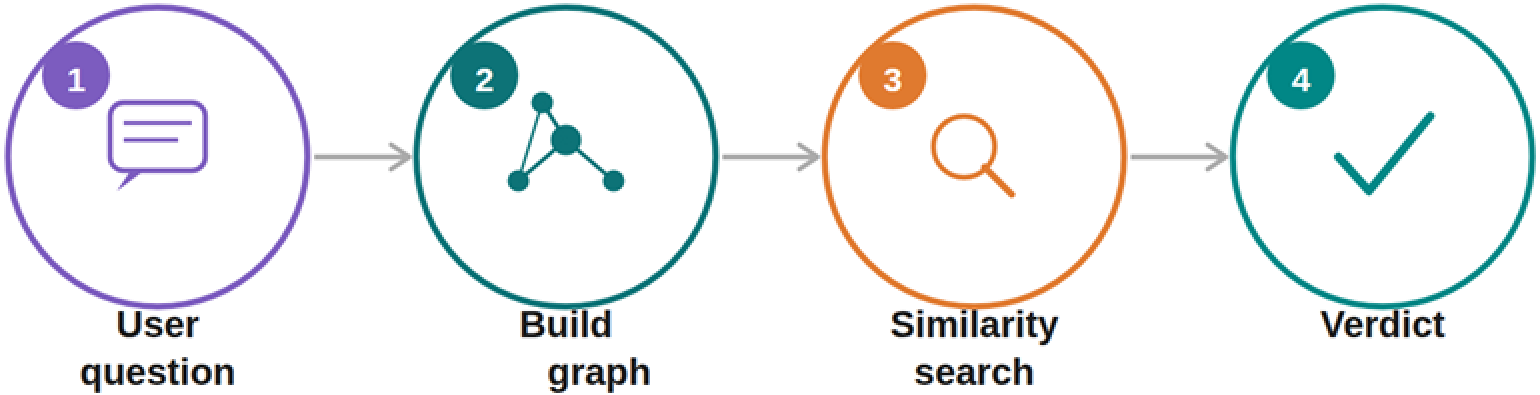
(**Claim**: “In 2020, the COVID-19 crisis resulted in a considerable drop in GDP, estimated around 4–5%.”)-> [:**MEASURES**] -> (**Value** {number: "4–5", **unit**: "%"})

3. Graph Creation — Process

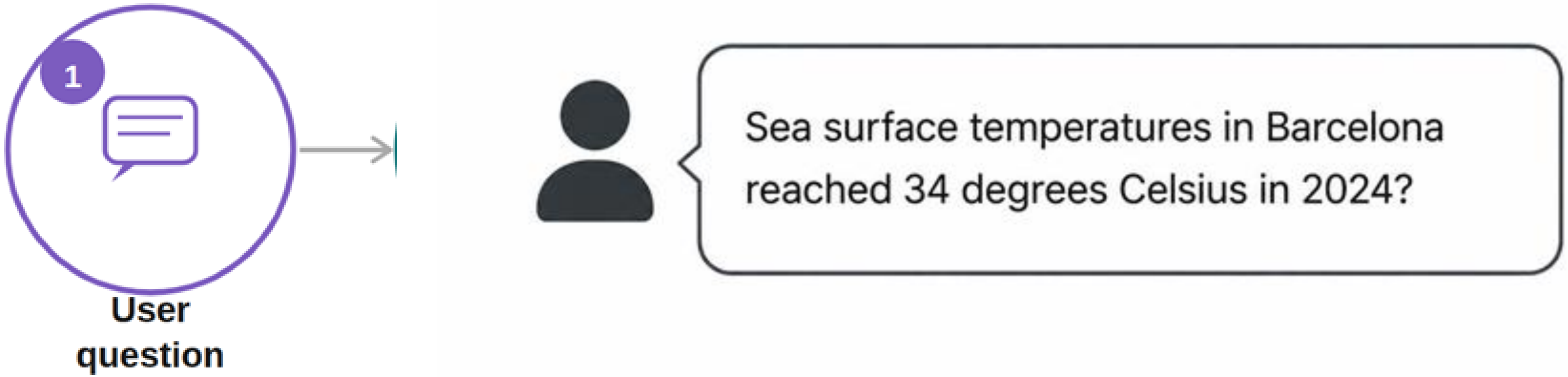
Create Neo4j graph



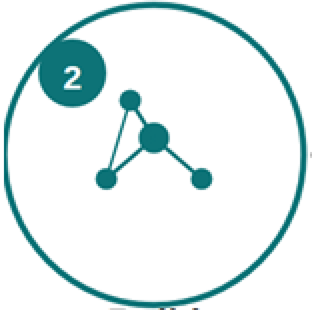
4. Task — Claim Verification



4. Task — Claim Verification



4. Task — Claim Verification



Build
graph

LLM

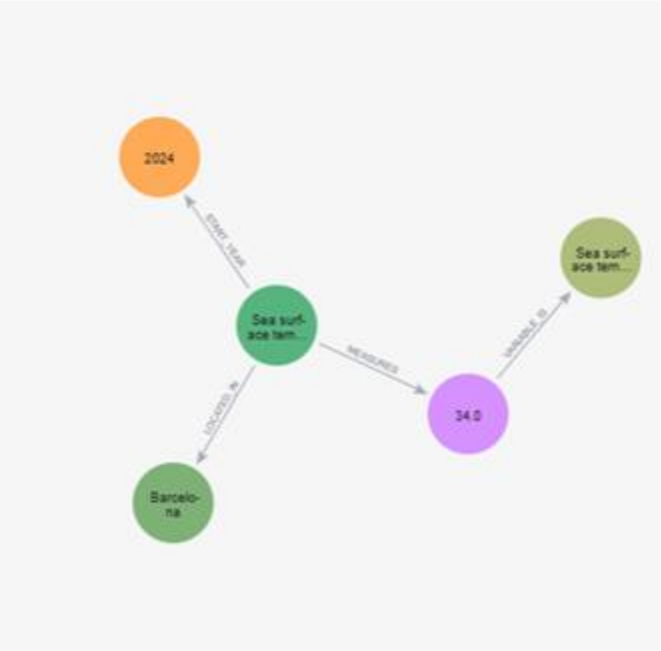
```
MERGE (c:Claim {
    text: "Sea surface temperatures in Barcelona reached 34 degrees Celsius"
})

MERGE (r:Region {
    name: "Barcelona"
})

MERGE (v:Value {
    number: 34.0,
    unit: "degrees Celsius"
})

MERGE (var:Variable {
    name: "Sea surface temperature",
    process: "Marine climate"
})

MERGE (c)-[:LOCATED_IN]->(r)
MERGE (c)-[:MEASURES]->(v)
```



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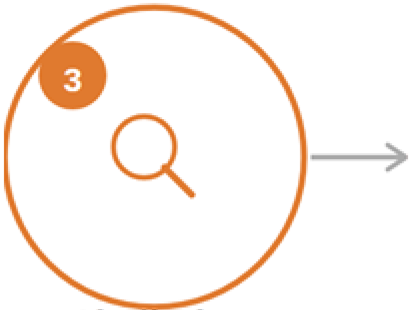
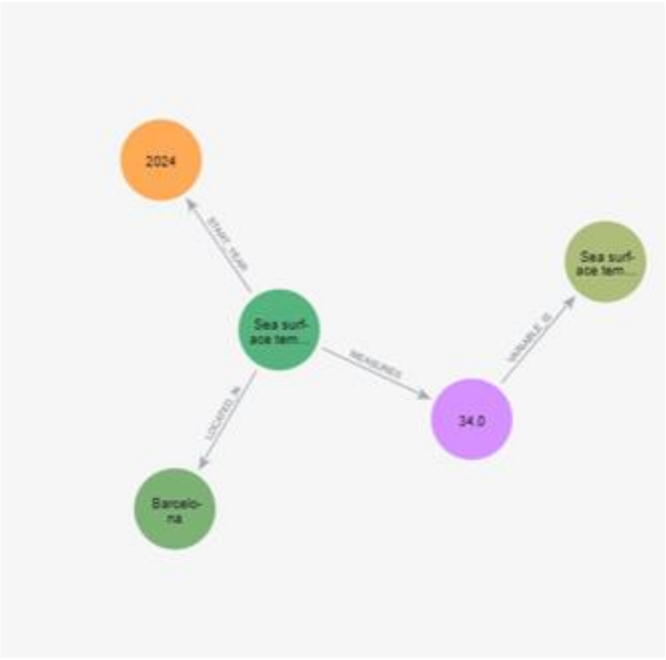


Similarity search

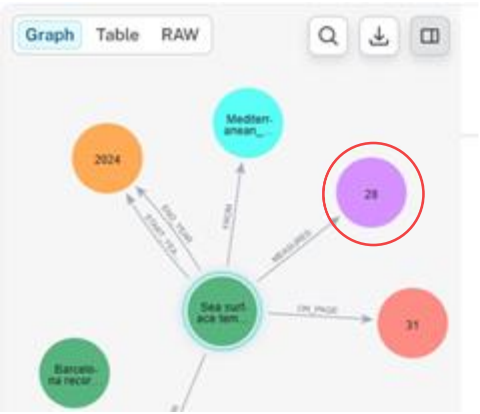
subgraph from ipcc reports

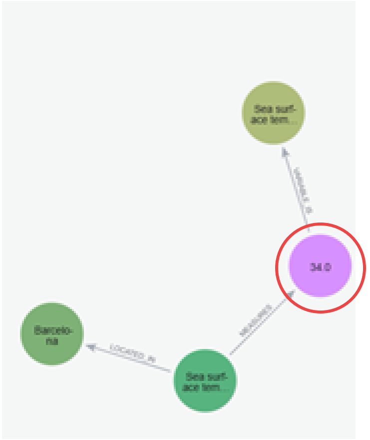


graph created from the claim

4. Task — Claim Verification

Verdict





Compare values

Example Comparison

✓ GROUND TRUTH

Sea surface temperatures in Barcelona during August 2024 reached anomalously high levels above 28 degrees Celsius

28.0 °C

Highest in instrumental record

VS

✗ CLAIM TO VERIFY

Sea surface temperatures in Barcelona reached 34 degrees Celsius in 2024

34.0 °C

Discrepancy: +6.0°C → FLAGGED

Summary

IPCC Source Material

Official climate reports as the ground truth for climate claims

Ontology Design

7-node schema (Claim, Document, Page, Value, Variable, Region, Time) with 8 relationship types.

Neo4j Knowledge Graph

Claims embedded by location, time, value & metric for semantic subgraph matching

Fact Checking Pipeline

Question → graph → similarity search → numerical comparison → verdict